Homework 4

Design and Analysis of Algorithms

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TASK 1:

```
MaxSubArraySum(A, n)
{ globalSum = A[1]
  MaxSum[1] = A[1]
  globalstart = 1  // Initalize the start of the
                   // maximum subarray.
  localstart = 1   // Initalize the Local start
                   // of the current subarray.

  for (i = 2 to n)
     if (MaxSum[i-1] + A[i] > A[i])
         MaxSum[i] = MaxSum[i-1] + A[i]
     else
         MaxSum[i] = A[i]
         localstart = i  // Update the local start of
                         // the current subarray

     If (globalSum < MaxSum[i])
         globalSum = MaxSum[i]
         globalEnd = i
         globalstart = localstart  // update the start
                                   // of the maximum
                                   // subarray
  return globalSum
}
```

TASK 2

MaxSum[1] = 2

MaxSum[2] = -2

[1, 2] = 2 + (-4) = -2 ✓
[2] = -4

MaxSum[3] = 3.

[1, 2, 3] = 2 + (-4) + 3 = 1
[2, 3] = (-4) + 3 = -1
[3] = 3 ✓

MaxSum[4] = 7

[1, 2, 3, 4] = 2 + (-4) + 3 + 4 = 5
[2, 3, 4] = (-4) + 3 + 4 = 3
[3, 4] = 3 + 4 = 7 ✓
[4] = 4

MaxSum(5) = 4

[1, 2, 3, 4, 5] = 2 + (-4) + 3 + 4 + (-3) = 2
[2, 3, 4, 5] = (-4) + 3 + 4 + (-3) = 0
[3, 4, 5] = 3 + 4 + (-3) = 4 ✓
[4, 5] = 4 + (-3) = 1
[5] = -3

MaxSum[6] = 9

[1, 2, 3, 4, 5, 6] = 2 + (−4) + 3 + 4 + (−3)
+ 5 = 7

[2, 3, 4, 5, 6] = (−4) + 3 + 4 + (−3) + 5
= 5

[3, 4, 5, 6] = (3) + 4 + (−3) + 5 = 9 ✓

[4, 5, 6] = 4 + (−3) + 5 = 6

[5, 6] = (−3) + 5 = 2

[6] = 5

MaxSum[7] = 4

[1, 2, 3, 4, 5, 6, 7] = 2 + (−4) + 3 + 4
+ (−3) + 5 + (−5) = 2

[2, 3, 4, 5, 6, 7] = (−4) + 3 + 4 + (−3)
+ (5 + (−5) = 0

[3, 4, 5, 6, 7] = 3 + 4 + (−3) + 5 + (−5) = 4 ✓

[4, 5, 6, 7] = 4 + (−3) + 5 + (−5) = 1

[5, 6, 7] = (−3) + 5 + (−5) = −3

[6, 7] = 5 + (−5) = 0

[7,] = −5

MaxSum[8] = 10

[1, 2, 3, 4, 5, 6, 7, 8] = 2 + (−4) + 3 + 4 + (−3)
+ 5 + (−5) + 6 = 8

[2, 3, 4, 5, 6, 7, 8] = (−4) + 3 + 4 + (−3)
+ 5 + (−5) + 6 = 6

[3, 4, 5, 6, 7, 8] = (3) + 4 + (−3) + 5
+ (−5) + 6 = 10 ✓

[4, 5, 6, 7, 8] = 4 + (−3) + 5 + (−5) + 6 = 7

[5, 6, 7, 8] = (−3) + 5 + (−5) + 6 = 3

[6, 7, 8] = 5 + (−5) + 6 = 6

[7, 8] = -5 + 6 = 1
[8] = 6

<u>MaxSum[9] = 9</u>

[1, 2, 3, 4, 5, 6, 7, 8, 9] = 2 + (-4) + 3 + 4
+ (-3) + 5 + (-5) + 6 + (-1) = 7
[2, 3, 4, 5, 6, 7, 8, 9] = (-4) + 3 + 4 +
(-3) + 5 + (-5) + 6 + (-1) = 5
[3, 4, 5, 6, 7, 8, 9] = 3 + 4 + (-3) + 5
+ (-5) + 6 + (-1) = 9
[4, 5, 6, 7, 8, 9] = 4 + (-3) + 5 + (-5)
+ 6 + (-1) = 6
[5, 6, 7, 8, 9] = (-3) + 5 + (-5) + 6
+ (-1) = 2
[6, 7, 8, 9] = 5 + (-5) + 6 + (-1) = 5
[7, 8, 9] = (-5) + 6 + (-1) = 0
[8, 9] = 6 + (-1) = 5
[9] = -1

TASK 3

MaxSum [1] = 2
MaxSum [2] = 2 + (−4) = −2

MaxSum [3] = 2 + (−4) = −2
Result is less than 0, so sum = 0
Sum + 3 = 3
MaxSum[3] = 3

MaxSum [4] = 2 + (−4) = −2
Result is less than 0, so sum = 0
Sum + 3 = 3
Sum + 4 = 7

MaxSum[4] = 7

MaxSum [5] = 2 + (−4) = −2
Result is less than 0, so sum = 0
Sum + 3 = 3
sum + 4 = 7
sum + (−3) = +4

MaxSum [5] = 4

MaxSum[6] = 2 + (−4) = −2
Sum is less than 0, so sum = 0
sum + 3 = 3
sum + 4 = 7
sum + (−3) = 4
sum + 5 = 9

MaxSum [6] = 9

$MaxSum[7] = 2 + (-4) = -2$

Sum less than 0, so sum = 0

$sum + 3 = 3$

$sum + 4 = 7$

$sum + (-3) = 4$

$sum + (5) = 9$

$sum + (-5) = 4$

$\underline{MaxSum[7] = 4}$

$MaxSum[8] = 2 + (-4) = -2$

Sum less than 0, so sum = 0

$sum + 3 = 3$

$sum + 4 = 7$

$sum + (-3) = 4$

$sum + 5 = 9$

$sum + (-5) = 4$

$sum + 6 = 10$

$\underline{MaxSum[8] = 10}$

$MaxSum[9] = 2 + (-4) = -2$

Sum less than 0, so sum = 0

$sum + 3 = 3$

$sum + 4 = 7$

$sum + (-3) = 4$

$sum + (5) = 9$

$sum + (-5) = 4$

$sum + 6 = 16$

$sum + (-7) = 9$

$\underline{MaxSum[9] = 9}$

TASK 4

We can only keep track of the current sum and the maximum sum seen so far, without using an additional array.

Pseudocode:-

```
maxSoFar = A[1]
maxEndingHere = A[1]

for i from 2 to n:
    maxEndingHere = max (A[i], maxEndingHere + A[i])

    maxSoFar = max (maxSoFar, maxEndingHere)

return maxSoFar
```